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COMMERCIAL ITEM DESCRIPTION

ROPE, FIBROUS (ARAMID)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This Commercial Item Description covers four-strand aramid fiber rope which is intended for use as mooring lines and floating dry dock tending lines.

2. SALIENT CHARACTERISTICS.

2.1 Description. The aramid fiber rope shall be fabricated from four strands right regular laid with a three-strand right regular laid polypropylene core. The center core shall consist of three strand polypropylene with a braded jacket of polyester yarn. The center core shall consist of three strand polypropylene with a braided jacket of alternating aramid and polyester yarns. The color of the finished rope shall be natural.

2.2 Braided strand jacket. The type of braid, number of strands, yarns per strand, and multipliers shall be in accordance with tables I and II. The jacket shall fit firmly around each aramid strand and polypropylene core when inspected with the rope in a relaxed state.

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FSC 4020

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TABLE I. Structural requirements for strand jacket.

Rope circumference [1/]		Type of braid	Number of strands	Pick count per mm (±10%)
millimeters (mm)	inches			
85	3⅜	Twill or diamond	24	0.40
89	3½	Twill or diamond	24	0.40
94	3¾	Twill or diamond	24	0.40
105	4⅛	Twill or diamond	24	0.40
120	4¾	Twill or diamond	24	0.40
		Twill or diamond	32	0.40
136	5⅜	Twill or diamond	24	0.40
		Twill or diamond	32	0.40
148	5⅞	Twill or diamond	24	0.40
		Twill or diamond	32	0.40
		Twill or diamond	36	0.40
		Twill or diamond	48	0.40
160	6¼	Twill or diamond	24	0.40
		Twill or diamond	32	0.40
		Twill or diamond	36	0.40
		Twill or diamond	48	0.40
193	7⅝	Twill or diamond	32	0.40
		Twill or diamond	36	0.40
		Twill or diamond	48	0.40
207	8 3/16	Twill or diamond	32	0.40
		Twill or diamond	36	0.40
		Twill or diamond	48	0.40

NOTE:

[1/] Measurements are approximate and included for information only.

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TABLE II. Structural requirements for core jacket.

Rope circumference [1/]		Number of strands
mm	inches	
19	¾	16 24
21	13/16	16 24
25	1	16 24
28	1 3/32	16 24
30	1 3/16	16 24
32	1¼	16 24
41	1⅝	24 32
48	1⅞	24 32
57	2¼	24 32
64	2½	24 32

NOTE:

[1/] Measurements are approximate and included for information only.

2.3 Fiber finish. An overlay finish for the purpose of reducing yarn-on-yarn abrasion and compression fatigue shall be provided. Manufacturer shall maintain a record of verifiable information attesting to the fact that marine overlay finish has been applied to the fibers. The overlay finish provided shall not violate any other requirements of this specification.

2.4 Physical properties. The finished rope shall conform to the physical properties specified in table III when tested in accordance with CI-1500A and CI-1500B.

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TABLE III. Physical properties.

Circumference (+5%/-10%)		Linear density (+5%/-10%)	Lay length maximum	Breaking strength	
mm	inches 1/	kg/100m	mm	kN	pounds 2/
85	3⅜	49	120	222	50,000
89	3½	55	127	267	60,000
94	3¾	62	133	311	70,000
105	4⅛	75	143	427	96,000
120	4¾	98	164	601	135,000
136	5⅜	128	184	801	180,000
148	5⅞	168	215	1,000	225,000
160	6¼	211	229	1,245	280,000
193	7⅝	259	270	1,557	350,000
207	8 3/16	312	290	1,868	420,000

NOTES:

1/ Inches are approximate and included for information only.

2/ Pounds are approximate and included for information only.

2.5 Elongation. The total elongation of the rope shall not exceed six percent at the minimum breaking strength value specified in table III. The load elongation curve, drawn autographically, shall not exhibit evidence of sudden changes in load applications greater than five percent of the load measured at the instant of change. Changes due to splice slippage shall not be considered in this determination.

2.6 Failure mechanism. The failure mechanism shall consist of sequential rupturing of the individual strands.

2.7 Lay length. The lay length shall not exceed the values as specified in table III.

2.8 Identification marker. The manufacturer shall identify each product by inserting a water resistant marker inside one strand in all sizes of ropes. The manufacturer's name, the year of manufacture, and type of fiber (aramid) shall be printed on the marker in bold, easy-to-read type. Italic or script type shall not be used. The printing shall not be affected by exposure to salt water or mineral oil.

2.9 Identification tag. In addition to the requirements specified in 2.8, each package unit shall have a ticket (identification tag) attached for identification purposes. The ticket shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information:

a. Stock number.
b. Nomenclature.
c. CID number.
d. Length.
e. Contract number and date.
f. Date of manufacture and year.
g. Contractor's name.

2.10 Spliceability. The finished rope shall be spliceable and shall not develop yarn displacement or strand hockles (distortion formed by back twist in the strand).

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2.11 Put up. Unless otherwise specified herein, the rope shall be provided in standard lengths of between 183 meters (600 feet) and 201 meters (660 feet) when measured in the relaxed condition and shall be furnished without knots or splices on non-returnable reels not larger than 2 meters (7 feet) in diameter. Multiple lengths shall be permitted but no piece shall be less than 183 meters (600 feet).

3. PRODUCT CONFORMANCE PROVISIONS.

3.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

3.2 Quality conformance inspection. A random sample of reels shall be selected from each lot offered to the government in accordance with ANSI/ASQ Z1.4 at inspection level II and acceptance quality limit (AQL) of 1.0.

3.3 End item visual examination. Each sample shall be examined for the following defects:

a. Defects in design, construction, or workmanship.
b. Any cut, kink, broken or loose ends, bulged strands.
c. Not of continuous length, contains knots, splices or otherwise secured to make a continuous length.
d. Any spot or stain clearly noticeable.
e. Ends not secured, whipped, or taped.
f. Length of unit more or less than specified (including permitted tolerances).
g. Length of unit less than marked on ticket.
h. Reels not in continuous length.
i. Reels improperly or not firmly wound, resulting in slippage during winding.

If any sample unit has one or more defects, the entire lot shall be rejected.

3.4 Testing of end item. The sample shall be tested as specified in 3.4.1 through 3.4.5. The physical and chemical values apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. If any sample unit has one or more defects, the entire lot shall be rejected.

3.4.1 Circumference. The measurement for circumference shall be made in accordance with CI-1500A.

3.4.2 Linear density. Linear density (kg/100m) shall be determined in accordance with CI-1500A.

3.4.3 Breaking strength. Breaking strength shall be determined in accordance with CI-1500B.

3.4.4 Elongation. Elongation shall be determined in accordance with CI-1500B.

3.4.5 Lay length. The length measurement shall be made in accordance with CI-1500A.

3.4.6 Failure mechanism. The failure mechanism shall be determined by testing one rope sample from each lot. The distance between the outer ends of the splices shall be 16 meters (50 feet) in length. The time between the first and second ruptures of the strands (warning time) shall be 10 seconds or more when tested in accordance with CI-1502. The sample shall meet the breaking-strength requirements of table III.

4. RESPONSIBILITY FOR COMPLIANCE. All items shall meet all requirements of the Commercial Item Description. The government will provide a Quality Assurance Representative to select the sample to be tested, witness all visual inspections, physical testing, and certify that each lot meets all requirements of this Commercial Item Description at the source. Sampling and inspection in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material. Failure to meet any of the requirements of this Commercial Item Description shall be cause for rejecting the entire lot. Retesting of the lot shall not be permitted.

5. PACKAGING. Preservation, packaging, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Source of documents.

6.1.1 The Cordage Institute. Cordage Institute publications are available online at www.ropecord.com.

6.1.2 American Society for Quality. ASQ standards are available online at www.asq.org.

6.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Circumference required (see 2.4).

c. When continuous lengths, other than standard, are required (see 2.11).

d. If purchaser will accept any unit which has been shortened or cut for test specimens if in complete compliance with this Commercial Item Description, and the reel shall be marked as to actual length.

6.3 National stock numbers (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

a. 4020-01-338-7006

b. 4020-01-338-7007

c. 4020-01-339-1571

d. 4020-01-339-5150

6.4 Key words.

Breaking strength

Elongation

Splice

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CQ

Preparing activity:
Navy – SH
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